

Elastic Net Regression

Nguyen Phu Nhan

Elastic Net regression is a regularized version of linear regression that combines the strengths of both Ridge regression and Lasso regression. Similar to ordinary linear regression, its objective is to model the relationship between a set of input variables and a continuous target variable. However, Elastic Net introduces both an L_1 penalty and an L_2 penalty in order to reduce model complexity, prevent overfitting, handle correlated features, and perform feature selection.

Suppose we are given a dataset consisting of n observations and p features. Let

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

be the feature matrix and

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

be the vector of target values.

As in linear regression, we absorb the bias term into the feature matrix by appending a column of 1's to $\mathbf{X}$. The prediction function can then be written as

$$\hat{\mathbf{y}} = \mathbf{X}\mathbf{w},$$

where $\mathbf{X} \in \mathbb{R}^{n \times p}$ is the feature matrix, $\mathbf{w} \in \mathbb{R}^p$ is the coefficient vector, and $\hat{\mathbf{y}} \in \mathbb{R}^n$ is the vector of predicted target values.

In ordinary linear regression, we minimize the mean squared error loss

$$L(\mathbf{w}) = \frac{1}{2n} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2.$$

Elastic Net regression modifies this objective function by adding both an L_1 penalty and an L_2 penalty:

$$L(\mathbf{w}) = \frac{1}{2n} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 + \lambda_1 \|\mathbf{w}\|_1 + \frac{\lambda_2}{2} \|\mathbf{w}\|_2^2,$$

where

$$\|\mathbf{w}\|_1 = \sum_{j=1}^p |w_j|$$

is the L_1 norm and

$$\|\mathbf{w}\|_2^2 = \sum_{j=1}^p w_j^2$$

is the squared L_2 norm.

Elastic Net is commonly parameterized using two hyperparameters: α and `l1_ratio`. We define

$$\lambda_1 = \alpha \cdot \text{l1_ratio}$$

and

$$\lambda_2 = \alpha \cdot (1 - \text{l1_ratio}).$$

Throughout the derivation, we use the shorthand notation λ_1 and λ_2 . Equivalently,

$$L(\mathbf{w}) = \frac{1}{2n} (\mathbf{X}\mathbf{w} - \mathbf{y})^\top (\mathbf{X}\mathbf{w} - \mathbf{y}) + \lambda_1 \sum_{j=1}^p |w_j| + \frac{\lambda_2}{2} \sum_{j=1}^p w_j^2.$$

The parameters λ_1 and λ_2 control the strength of regularization.

- If `l1_ratio` = 1, Elastic Net reduces to Lasso regression.
- If `l1_ratio` = 0, Elastic Net reduces to Ridge regression.
- If $\alpha = 0$, the model reduces to ordinary linear regression.
- Increasing α increases the overall strength of regularization.
- Larger values of `l1_ratio` place more emphasis on feature selection through the L_1 penalty.

- Smaller values of `l1_ratio` place more emphasis on coefficient shrinkage through the L_2 penalty.

The training objective is therefore

$$\mathbf{w}^* = \underset{\mathbf{w}}{\operatorname{argmin}} \left[\frac{1}{2n} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 + \lambda_1 \|\mathbf{w}\|_1 + \frac{\lambda_2}{2} \|\mathbf{w}\|_2^2 \right].$$

Unlike ordinary linear regression and Ridge regression, there is generally no closed-form solution because of the non-differentiability of the absolute value function at zero. Consequently, Elastic Net is typically optimized using coordinate descent.

For a coefficient w_j , define the partial residual

$$r_i^{(j)} = y_i - \sum_{k \neq j} x_{ik} w_k.$$

Substituting this expression into the objective function yields

$$L(w_j) = \frac{1}{2n} \sum_{i=1}^n \left(r_i^{(j)} - x_{ij} w_j \right)^2 + \lambda_1 |w_j| + \frac{\lambda_2}{2} w_j^2.$$

Expanding the squared term gives

$$L(w_j) = \frac{1}{2n} \sum_{i=1}^n \left[(r_i^{(j)})^2 - 2x_{ij} r_i^{(j)} w_j + x_{ij}^2 w_j^2 \right] + \lambda_1 |w_j| + \frac{\lambda_2}{2} w_j^2.$$

Define

$$\rho_j = \sum_{i=1}^n x_{ij} r_i^{(j)}$$

and

$$z_j = \sum_{i=1}^n x_{ij}^2.$$

Substituting these quantities into the objective function yields

$$L(w_j) = \frac{z_j}{2n} w_j^2 - \frac{\rho_j}{n} w_j + \lambda_1 |w_j| + \frac{\lambda_2}{2} w_j^2 + C,$$

where

$$C = \frac{1}{2n} \sum_{i=1}^n (r_i^{(j)})^2$$

is a constant independent of w_j .

Ignoring constants, we minimize

$$L(w_j) = \left(\frac{z_j}{2n} + \frac{\lambda_2}{2} \right) w_j^2 - \frac{\rho_j}{n} w_j + \lambda_1 |w_j|.$$

Because the absolute value function is piecewise-defined, we consider three cases.

Case 1: $w_j > 0$ In this case,

$$|w_j| = w_j.$$

The objective becomes

$$L(w_j) = \left(\frac{z_j}{2n} + \frac{\lambda_2}{2} \right) w_j^2 - \frac{\rho_j}{n} w_j + \lambda_1 w_j.$$

Differentiating with respect to w_j gives

$$\frac{dL}{dw_j} = \left(\frac{z_j}{n} + \lambda_2 \right) w_j - \frac{\rho_j}{n} + \lambda_1.$$

Setting the derivative equal to zero,

$$(z_j + n\lambda_2) w_j = \rho_j - n\lambda_1.$$

Therefore,

$$w_j = \frac{\rho_j - n\lambda_1}{z_j + n\lambda_2}.$$

This solution is valid only when

$$\rho_j > n\lambda_1.$$

Case 2: $w_j < 0$ In this case,

$$|w_j| = -w_j.$$

The objective function becomes

$$L(w_j) = \left(\frac{z_j}{2n} + \frac{\lambda_2}{2} \right) w_j^2 - \frac{\rho_j}{n} w_j - \lambda_1 w_j.$$

Differentiating,

$$\frac{dL}{dw_j} = \left(\frac{z_j}{n} + \lambda_2 \right) w_j - \frac{\rho_j}{n} - \lambda_1.$$

Setting the derivative equal to zero,

$$(z_j + n\lambda_2) w_j = \rho_j + n\lambda_1.$$

Hence,

$$w_j = \frac{\rho_j + n\lambda_1}{z_j + n\lambda_2}.$$

This solution is valid only when

$$\rho_j < -n\lambda_1.$$

Case 3: $w_j = 0$ If

$$-n\lambda_1 \leq \rho_j \leq n\lambda_1,$$

the minimum occurs at

$$w_j = 0.$$

Combining all three cases, we obtain

$$w_j = \begin{cases} \frac{\rho_j - n\lambda_1}{z_j + n\lambda_2}, & \rho_j > n\lambda_1, \\ 0, & |\rho_j| \leq n\lambda_1, \\ \frac{\rho_j + n\lambda_1}{z_j + n\lambda_2}, & \rho_j < -n\lambda_1. \end{cases}$$

This piecewise expression can be written compactly using the soft-thresholding operator

$$S(\rho, \lambda) = \begin{cases} \rho - \lambda, & \rho > \lambda, \\ 0, & |\rho| \leq \lambda, \\ \rho + \lambda, & \rho < -\lambda. \end{cases}$$

Substituting

$$\lambda_1 = \alpha \cdot \text{ll_ratio}$$

and

$$\lambda_2 = \alpha \cdot (1 - \text{ll_ratio}),$$

the coordinate descent update becomes

$$w_j = \frac{S(\rho_j, n\alpha \cdot \text{ll_ratio})}{z_j + n\alpha (1 - \text{ll_ratio})}.$$

The soft-thresholding operator shrinks coefficients toward zero and sets small coefficients exactly equal to zero, while the Ridge component further shrinks all coefficients through the denominator.

Consequently, Elastic Net performs both feature selection and coefficient shrinkage simultaneously. This makes it particularly useful when the number of features is large, when many features are irrelevant, or when predictors are highly correlated.